

COURSE OUTLINE

Course ID	Course Title	Credits
AI301DE01	MÁY HỌC	03
	Machine Learning	

(Applied to Semester: 1A, Academic year: 2023-2024,
Decision No. 2614/QĐ-ĐHHS dated 30/10/2023)

A. Course Specifications:

Periods				Periods in classroom		
Total periods*	Lecture/Seminar	Activity	Self-study periods	Lecture room	Lab room	Fieldwork
(1)	(2)	(3)	(4)	(5)	(6)	(7)
60	30	30	90	30	30	0

$$(1) = (2) + (3) = (5) + (6) + (7)$$

B. Other related Subjects:

Other related Subjects	Course ID	Course title
Prerequisites:		
1.	SW103D or IT208D or AI202DV01 or AI207DV01	Object-oriented Programming or System Object-oriented Analysis, Design and Development or Object-oriented Programming in Python or System Object-oriented Analysis, Design and Construction with Python
2.	IT201D or IT209D	Database or Database Design and Management

C. Course Description:

Machine learning has been successfully applied to many different fields such as automatic control of cars and robots, natural language processing, image recognition, health sciences, biology and data mining. This course introduces fundamental concepts and methods for learning from data for computational data analysis, including pattern recognition, prediction, and visualization. Therefore, this subject includes supervised learning, unsupervised learning, and reinforcement learning. This course mainly focuses on applying machine learning techniques to various problems.

D. Course Objectives:

No.	Course Objectives
O1	Get familiar with data and visualization tools

O2	Build machine learning problems for various applications
O3	Understand the basic theory of machine learning
O4	Apply machine learning algorithms for problem solving

E. Learning Outcomes:

No.	Learning Outcomes	Contribute to the program (AI)
O4	CLO1 Set up and solve typical machine learning problems, either hands-on or by using established computer simulation tools	PLO 2
O3	CLO2 Decide on the right machine learning method/algorithm for the type of data learning problem, i.e., know about their most important weaknesses and advantages.	PLO 2
O3	CLO3 Decide how to present the data to facilitate learning the data	PLO 2
O2	CLO4 Recognize typical effects of bad initialization and parameter selection and suggest ways to improve results	PLO 2
O1	CLO5 Describe how, and why, machine learning and natural computing methods work, explaining principles and presenting examples.	PLO 2

F. Instructional Modes:

	Room	Periods
1	Lecture room	30
2	Lab room	30
	Total	60

Lectures will be combined with in-class discussion and group presentation with some requirements: student must read book, do homework, practice and prepare lesson before going to the class.

No.	Instructional Modes	Description	Number of periods
1	Lecture	Teaching with slides in class, combining a number of methods such as conversation, lecture, and experiment (on a simulation model in class)	30
2	Group work	- Students perform weekly exercises/practices assigned by the instructor. - If any exercise/practice cannot be done, students should discuss with the instructor for guidance	30

G. Textbooks and teaching aids:

1. Textbooks:

[1] Bashier, Eihab Bashier Mohammed_ Khan, Muhammad Badruddin_ Mohammed, Mohssen - *Machine learning_ algorithms and applications*, 2017, CRC Press.

[2] Tom M. Mitchell - *Machine Learning*, 1997, MGH.

[3] Slides provided by lecturer

2. References:

3. Software Apps: Microsoft Visual Studio , R, WEKA

H. Assessment Methods (Requirements for Completion of the Course):

1. Description of learning outcomes assessment

a) Quiz:

15-minute multiple-choice test at the beginning of theory lessons. Test content is based on previous lesson knowledge through multiple-choice questions and short answered questions. Students are not allowed to use the materials. Questions are given by the theory lecturer. This test accounts for 10% of the total subject score..

b) Practical exercises:

Every week, the instructor will give requirements and exercises. Students must do their INDIVIDUAL work and submit it according to the deadlines set by the instructor. This test accounts for 30% of the total subject score.

c) Tests

There are two routine tests. Testing time is determined by the lecturer. The test is multiple choice or essay or includes both multiple choice and essay parts. Duration is 60 minutes. Students ARE allowed to use the materials. This test accounts for 20% of the total subject score.

d) Final Project:

Students carry out the topic according to the teacher's request, working in groups of maximum 3 students. Submit your assignment at the last class session. This part accounts for 40% of the total score of the subject.

2. Summary of learning outcomes assessment

* For main/extra semester:

<i>Components</i>	<i>Duration</i>	<i>Assessment Forms</i>	<i>Percentage</i>	<i>Schedule</i>	<i>CLOs</i>
<i>Quiz</i>	<i>10' quiz</i>	<i>8-10 quizzes</i>	<i>10%</i>	<i>Week 2 -14</i>	<i>4,5</i>
<i>Practical exercises</i>		<i>Individual work</i>	<i>30%</i>	<i>Week 4, 8, and 12</i>	<i>1-5</i>
<i>Tests</i>	<i>60'</i>	<i>Two routine tests. Multiple choice or essay questions or include both multiple choice and essay parts</i>	<i>20%</i>		<i>2-5</i>
<i>Final project</i>		<i>Group of up to 3 students.</i>	<i>40%</i>	<i>Week 15</i>	<i>1-5</i>
<i>Total</i>			<i>100%</i>		

I. Academic integrity

Academic integrity is a fundamental value that affects the quality of teaching, learning, and research at a university. To ensure the maintenance of academic integrity at Hoa Sen University, students are required to:

1. Work independently on individual assignments: Collaborating on individual assignments is considered cheating.

2. Avoid plagiarism: Plagiarism is an act of fraud that involves the use of ideas or words of another person without proper attribution. Students will be accused of plagiarism if they:

a. Copy in their work one or more sentences from another person without proper citation.

b. Rephrase, paraphrase, or translate another person's ideas or words without proper attribution.

c. Reuse their own assignments, in whole or in part, and submit them for another class.

d. Work responsibly within a working group.

3. In cooperative group assignments, all students are required to stay on task and contribute equally to the projects. Group reports should clearly state the contribution of each group member. Any acts of academic dishonesty will result in a grade of zero for the task at hand and/or immediate failure of the course, depending on the seriousness of the fraud. Please consult Hoa Sen University's Policy on Plagiarism at <http://thuvien.hoasen.edu.vn/chinh-sach-phong-tranh-dao-van>

To ensure the maintenance of academic integrity, the university asks that students report cases of academic dishonesty to the teacher and/or the Dean. The names of those students will be kept anonymous.

J. Teaching Staff:

No.	Lecturer's name	Email, Phone number, Office location	Office hours	Position
1	Lê Thanh Tùng	Tung.LeThanh@hoasen.edu.vn		Lecturer
2	Trang Hồng Sơn	Son.TrangHong@hoasen.edu.vn		Lecturer

K. Outline of Topics to be covered:

- For main/extra semester:

Week	Lecture Content	Reference Material	Work to be completed by students	CLOs
1/1	Course Introduction Introduction: Machine Learning	[1].Chapter 1	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 5
SECTION I SUPERVISED LEARNING ALGORITHMS				
2/2	Classifiers and Error Rates			CLO 1
3/3	Decision Trees • Introduction to Decision Trees • Entropy • Attribute Selection Measure • Implementation	[1].Chapter 2	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 4

4/4	Rule-Based Classifiers • Introduction to Rule-Based Classifiers • Sequential Covering Algorithm • Algorithm • Visualization • Ripper	[1].Chapter 3	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 4
5/5	Bayesian Classifiers, Naive Bayes • Introduction to Bayesian Classifiers • Prior Probability • Likelihood • Laplace Estimator • Posterior Probability • Implementation	[1].Chapter 4	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 4
6/6	The k-Nearest Neighbors Classifiers • Introduction to the k-Nearest • Example • Implementation k-Nearest Neighbors	[1].Chapter 5	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 3
7/7	Neural networks • Perceptron Neural Network • Multilayer Perceptron Networks • The Backpropagation Algorithm • Implementation of Neural Networks	[1].Chapter 6	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 1
8/8	Linear Discriminant Analysis (LDA) • Introduction to LDA • Example • Implementation LDA	[1].Chapter 7	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 3

9/9	Support vector machines (SVMs) • Introduction to SVM • Definition of the Problem • Implementation SVM	[1].Chapter 8	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 3
SECTION II UNSUPERVISED LEARNING ALGORITHMS				
10/10	k-Means Clustering • Introduction to k-Means Clustering • Description of the Method • The k-Means Clustering Algorithm • Implementation of k-Means	[1].Chapter 9	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
11/11	Gaussian Mixture Model • Introduction • Learning the Concept by Example	[1].Chapter 10	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
12/12	Hidden Markov Model (HMM) • Introduction to HMM • Example • Implementation HMM	[1].Chapter 11	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 2
13/13	Principal Component Analysis (PCA) • Introduction to PCA • Description of the Problem • The Idea behind the PCA ◦ The SVD and Dimensionality Reduction • PCA Implementation	[1].Chapter 12	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 1

14/14	Reinforcement learning • Introduction to Reinforcement learning • Example • Implementation of Reinforcement learning	[1].Chapter 13	• Read the materials before going to theory class. • Complete assignments in practical class and submit to Mlearning	CLO 1
15/15	Final project grading		• Complete all project tasks and submit to Mlearning	

Date: 15/9/2023

Written by

(Signature and the full name)



Lê Thanh Tùng

Date: 15/9/2023

Program Chair

(Signature and the full name)



Lê Thanh Tùng

Date: 15/9/2023

Dean/ Vice Dean

(Signature and the full name)



Lê Đình Phong

